HU Extension Assignment 04 E89 Advanced Deep Learning

Posted on: 09/25/2026 Due by 1:00 PM EST on Saturday, 10/03/2026

It is our advice that you work with Anaconda and use Conda to create virtual environments. If you know what you are doing, choose your environment as you please. The virtual environment for this course should have Python 3.10 or higher and PyTorch 2.9 or higher. Most code in our examples will work with lower versions of PyTorch, though.
You should have a local installation of PyTorch. We will not penalize you if you do not.

Problem 1. We saw a simple CNN notebook (1_pytorch_basic_cnn_digits.ipynb) where digits from Sklearn were used and a classification model was created. You will expand on that work and instead of sklearn dataset, use the MNIST dataset from torchvision where images are of size 28 x 28. Build model with torchvision Dataset. Next draw a few numbers in your own handwriting and see if the model can detect it, just inference. Do not over do this.
(30%)

Problem 2. As part of the 2_pytorch_CNN_Classify_flowers.ipynb we saw the model was trained to a 75% accuracy. Change the hyper parameters – add new layers, change activation functions, experiment with different kernel sizes, experiment pooling layers types and see if you can increase the accuracy of the model. You should add 2 more convolutional layers (blocks) and make the network even deeper, also add one additional linear layer with RELU activation. Document your experiments in detail, don't overdo this and ensure to not overfit. Do not worry about doing any Data augmentation techniques at this point.
(30%)

Problem 3. In the 3_Visualizing_Flower_CNN_Filters_PyTorch.ipynb we did an introspection of how a Daisy image moves through a CNN and what feature maps are created at every block. Now, we have increased the depth of network in problem 2 above, we added 2 more convolutional layers (blocks) also added one linear layer with Relu activation as well. Make all these changs here in this notebook as well and load our new weights. And make sure we can still visualize the daisy. Notice our Dataset has Roses and tulips and other flowers, pick any two other flowers types and see how their feature maps look at last block. Also generate grad-CAM heatmap for both flowers and see if it truly highlights what makes up that image as a flower.
(40%)

Formatting Requirements:

Please name your notebook as `e89-LastName-FirstName-HW04.ipynb`

Solution of this assignment should be submitted as one Jupyter notebook (ipynb file), as well as an HTML image of that notebook. To generate the HTML image of you notebook, use the following command:

```
$ jupyter nbconvert e89-LastName-FirstName-HW04.ipynb --to html
Or
$ jupyter nbconvert --to html e89-LastName-FirstName-HW04.ipynb
```

Please provide meaningful descriptions of all actions in all cells of your notebook. Include those descriptions either as Python comments (#) or as the contents of the adjacent markdown cells.

If your notebook(s) contain(s) excessively long outputs, please copy a meaningful and illustrative number of initial and/or final lines and paste those in a markdown (comment) cell. Then, delete the long output(s). Thank you.